Q.NO	QUESTION	MARKS	SECTION	UNIT
1	Define Data Warehouse and Its Features in Detail?	8	Section-I	1
2	Distinguish between Operational Data base (OLTP) and Data Warehouse?	8	Section-I	1
3	Illustrate Data warehouse Architecture and its components?	8	Section-I	1
4	a) Discuss about multi- dimension table and data cube technology ? (4M) b) Explain clearly about Meta data repository and data warehouse models? (4M)	8	Section-I	1
5	Explain about Data modelling and types of Schemas With diagram?	8	Section-I	1
6	Draw a neat diagram for three-tier architecture and elaborate it in detail?	8	Section-I	1
7	a) What are the types of OLAP operations, explain with example? (4M) b) Discuss about Dimension and Fact table? (4M)	8	Section-I	1
8	Difference between types of OLAP servers -ROLAP vs MOLAP vs HOLAP?	8	Section-I	1
9	Explain about a) Data marts, Meta data (4M) c) Enterprise warehouse, virtual warehouse (4M)	8	Section-I	1
10	Discuss in detail about the characteristics of Data warehouse?	8	Section-I	1
11	Define Data Mining and Explain about Multi- Dimensional view of Data Mining?	8	Section-II	2
12	Define Data Mining and its Functionalities in detail?	8	Section-II	2
13	Illustrate about KDD with diagram? (4M)	8	Section-II	2
14	a) What are the major issues in data mining? (4M) b) Discuss about concept hierarchy generation? (4M)	8	Section-II	2
15	What is Data pre-processing and explain various pre-processing techniques?	8	Section-II	2
16	How to handle Noisy data and missing data in data cleaning?	8	Section-II	2
17	a) Define co-relation and co-variance? (4M) b) Discuss about PCA and Attribute subset selection? (4M)	8	Section-II	2
18	Describe about Data Integration briefly?	8	Section-II	2
19	Elaborate about Data Reduction and its strategies used in data mining.	8	Section-II	2
20	What Kinds of Patterns Can Be Mined in Data mining?	8	Section-II	2
21	Explain what is frequent pattern mining . How is it relevant in the context of market basket analysis? Provide an example to illustrate your answer.	8	Section-III	3
22	Discuss the importance of support and confidence measures in association rule mining. How are these measures calculated, and what do they signify?	8	Section-III	3
23	Describe the concept of market basket analysis. How does it help retailers in understanding customer purchasing behavior?	8	Section-III	3
24	What are the Association rules and its algorithms related?	8	Section-III	3
25	Describe the Apriori algorithm step-by-step. Include how it generates frequent itemsets and discovers association rules. Transactional data for an *AllElectronics* branch. *TID* — *List of item_IDs* T100 — I1, I2, I5 T200 — I2, I4 T300 — I2, I3 T400 — I1, I2, I4 T500 — I1, I3 T600 — I2, I3 T700 — I1, I3 T800 — I1, I2, I3, I5 T900 — I1, I2, I3	8	Section-III	3
26	Discuss strategies for improving the efficiency of the Apriori algorithm.	8	Section-III	3
27	Compare the Apriori algorithm with alternative approaches such as FP-Growth. What are the advantages and disadvantages of each approach in terms of efficiency and scalability?	8	Section-III	3

28	Elaborate about the FP-growth for this below table. Transactional data for an *AllElectronics* branch. TID — List of item_IDs T100 — I1, I2, I5 T200 — I2, I4 T300 — I2, I3 T400 — I1, I2, I4 T500 — I1, I3 T600 — I2, I3 T700 — I1, I3 T800 — I1, I2, I3, I5 T900 — I1, I2, I3	8	Section-III	3
29	Explain finding frequent item sets without candidate generation with suitable example?	8	Section-III	3
30	Discuss a short note on : a) Support count and Confidence . (4M) b)Market basket Analysis for frequent pattern mining. (4M)	8	Section-III	3
31	Define classification and prediction in the context of data mining. How do they differ from each other?	8	Section-IV	4
32	Describe the decision tree induction process. How does a decision tree algorithm construct a tree from training data? Class-Labeled Training Tuples from the *AllElectronics* Customer Database RID, age, income, student, credit_rating, Class: buys_computer 1, youth, high, no, fair, no 2, youth, high, no, excellent, no 3, middle_aged, high, no, fair, yes 4, senior, medium, no, fair, yes 5, senior, low, yes, fair, yes 6, senior, low, yes, excellent, no 7, middle_aged, low, yes, excellent, yes 8, youth, medium, no, fair, no 9, youth, low, yes, fair, yes 10, senior, medium, yes, fair, yes 11, youth, medium, yes, excellent, yes 12, middle_aged, medium, no, excellent, yes 13, middle_aged, high, yes, fair, yes 14, senior, medium, no, excellent, no	8	Section-IV	4
33	a) What is Classification? Explain the two classification models with example? (4M) b) Discuss about IF-THEN-ELSE Rule-based classification with example? (4M)	8	Section-IV	4
34	Explain the decision tree induction algorithm with appropriate examples. What is overfitting, and how can it be prevented for decision trees?	8	Section-IV	4
35	Explain how IF-THEN rules are used in rule-based classification. Provide an example of a rule-based classifier and discuss how rules are generated and applied.	8	Section-IV	4
36	Illustrate about Navie Bayes Classification with Bayes theorem?	8	Section-IV	4
37	Explain the K-Nearest Neighbour (K-NN) classification algorithm.	8	Section-IV	4
38	Discuss about Tree Pruning? a) Pre-pruning (4M) b)Post- pruning. (4M)	8	Section-IV	4
39	How will you solve classification problem using Bayesian belief networks?	8	Section-IV	4
40	Discuss about Attribute selection measure “Information Gain”? Class-Labeled Training Tuples from the *AllElectronics* Customer Database RID, age, income, student, credit_rating, Class: buys_computer 1, youth, high, no, fair, no 2, youth, high, no, excellent, no 3, middle_aged, high, no, fair, yes 4, senior, medium, no, fair, yes 5, senior, low, yes, fair, yes 6, senior, low, yes, excellent, no 7, middle_aged, low, yes, excellent, yes 8, youth, medium, no, fair, no 9, youth, low, yes, fair, yes 10, senior, medium, yes, fair, yes 11, youth, medium, yes, excellent, yes 12, middle_aged, medium, no, excellent, yes 13, middle_aged, high, yes, fair, yes 14, senior, medium, no, excellent, no	8	Section-IV	4
41	What is Cluster analysis and types of approaches seen ?	8	Section-V	5
42	Describe the basic principles of partitioning clustering methods. What are the advantages and limitations of algorithms like K- means and K-medoids?	8	Section-V	5

43	Explain the types of data structures and types of data commonly used in cluster analysis?	8	Section-V	5
44	Explain how hierarchical clustering methods construct dendrograms. What criteria are used to decide the merging or splitting of clusters in agglomerative and divisive approaches?	8	Section-V	5
45	Define density-based clustering methods such as DBSCAN. How do these methods identify clusters based on local density variations?	8	Section-V	5
46	Define cluster Analysis and their properties?	8	Section-V	5
47	a) Define outlier analysis and its detection methods? (4M) b) What are the types of outliers ? (4M)	8	Section-V	5
48	Define outlier analysis in the context of clustering. How do outliers impact clustering results, and what techniques are used to detect and handle outliers?	8	Section-V	5
49	Illustrate a short note on the major clustering methods : a. Partitioning Methods (4M) b. Hierarchical Methods (4M)	8	Section-V	5
50	Discuss briefly about DIANA and AGNES methods with example?	8	Section-V	5